

TITLE: Concise guidelines and complementary checklists for improving research reliability and reproducibility

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Standfirst

Checklists to address reproducibility shortfalls have proliferated but tend to be discipline- or research stage-specific and lack context or learning opportunities. We offer an alternative: one page, every stage, with a reason for each recommendation.

Keywords: checklist, reproducibility, transparency, research reliability, credibility, replicability, guidelines, R package

Introduction

Building literacy and skills to enhance research reliability and reproducibility is often hindered by lack of institutional support or training and otherwise takes second stage to maximizing the quantity and “novelty” of research outputs. The consequences of this have been repeatedly highlighted in the literature, across many, if not most scientific disciplines [1–4]: a) insufficient documentation (studies lacking data, scripts, context, i.e. complex analytical processing decisions, etc.), resulting in an inability to identify questionable research practices and thus measure bias; b) inadequate study design, including insufficient replication or inappropriate statistical analyses limiting the reliability of conclusions; c) an overreliance on binary criteria (i.e. $P < 0.05$), combined with inadequate reporting of effect sizes, confidence intervals, and uncertainty around estimates; d) poor differentiation across the nexus of exploratory (hypothesis generating) and confirmatory (hypothesis testing) research, resulting in unreliable statistical interpretations; d) a body of literature that is severely biased and misrepresents the full scope of research being conducted [5].

The gaps

Among the myriad resources available to researchers to strengthen or evaluate the reliability and reproducibility of outputs, checklists have emerged as popular tools. Checklists offer an accessible means to standardize reporting and evaluation, asking binary or scaled questions about procedures within the research lifecycle. Checklists have been developed across a range of applications [e.g. 6–9], generally tailored to specific disciplines [e.g. 10] and study designs [e.g. clinical trials; 11,12], or research life cycle phases such as preregistration (<https://www.cos.io/initiatives/prereg>) or archiving data [13] and code [14]. A few are more comprehensive and include interactive dashboards [e.g. 9], but remain limited in scope and

prescriptive in nature. Specifically, they aren't designed to help the user understand the problem being addressed by the checkbox. Thus, we note three shortfalls in the current landscape of checklists: a) to our knowledge, none are sufficiently discipline-agnostic (within the sciences); (b) none address the full research life cycle; and c) while they contribute to evaluating reporting, they fall short on supporting critical interpretation.

The solution

To address these shortfalls, we present a resource, entitled “Concise Guidelines for Producing Reliable, Reproducible Research”, that attempts to fill the gap between checklists and the well-developed literature that underpins what these checklists are attempting to evaluate. We anticipate it being especially useful for graduate students and early career researchers as it synthesizes common issues across the sciences and links out to core papers and resources through a shared public library (https://www.zotero.org/groups/6487727/concise_guidelines_for_reliable_research/library).

The need for such a resource became clear to us through the work we have been engaged in to enhance Open Science practices at the University of British Columbia's Okanagan campus [15] and beyond [16], and reflects discussions with graduate students and early career researchers through teaching, supervision, and project collaboration.

In conjunction with a companion R package (called “rrchecklist” for “reliable research” checklist [17]) that provides fillable checklists in HTML or PDF formats, we provide a proof-of-concept that merges the contextual resource with a checklist. Combined with the Concise Guidelines document, this resource enhances the ability of researchers to apply a critical lens to their interpretation of research outputs (their own and others) while directing users to a concise library of foundational resources for deeper learning.

69 **What it does**

70 The resource adopts a full research lifecycle approach, starting with the evaluation of existing
71 published research, enabling identification of the best available evidence to inform a new
72 research project. It then builds on this, mapping the concepts used in this evaluation to one's own
73 study design considerations, through to one's own reporting of outcomes. We see this as an
74 important feature; academia trains aspiring researchers well to be critical of others' works, but is
75 less effective at encouraging the application of a similarly critical lens to one's own work.
76 Another key feature of the resource is being prescriptive in what to consider, but not in how to
77 consider it, i.e. one should look for hallmarks of poor study design and selective reporting, one
78 should be able to distinguish between exploratory and confirmatory studies and appropriate
79 reporting in each, etc. As such, and when partnered with a checklist that identifies specific
80 criteria to look for in evaluating study design, the resource provides valuable informational
81 context. It encourages the user to recall their own foundational training around study design,
82 interpretation, and reporting – thereby resurrecting considerations that are otherwise often
83 abandoned in favour of convention (e.g. not considering statistical power, emphasizing statistical
84 significance over real world significance).

85 In its treatment of study design and statistical concepts, the resource implies a frequentist rather
86 than Bayesian framework. We consider this more broadly applicable for most researchers –
87 especially those at early career stages. We would welcome the development of a Bayesian
88 version of the guidelines.

89 The resource also addresses ethical community engagement and encourages collaboration with
90 those potentially impacted by the research [18], highlighting the importance of adhering to
91 community-specific protocols. Additionally, it addresses key aspects of research integrity and

scholarly communication, ensuring due credit and maximizing discoverability, accessibility, and reach.

What it does not do

While the resource provides links to a few key references that serve as foundations for the user to build off, it does not provide a comprehensive rationale or an extended list of references for any of its recommendations or statements.

The resource does not provide introductory or background material on the reproducibility crisis, study design, or statistics. Rather, it assumes the user has sufficient introductory knowledge of these topics such that most, if not all, of the terminology and statements are familiar to them.

Conclusion

Both the guidelines document and the checklist package are meant to be adapted and enhanced; in spite of an attempt to create a discipline-agnostic resource for the sciences, we recognize limitations resulting from our own experiences and anticipate that we have used language and references that will not resonate with all potential audiences. Releasing the guidelines document in multiple formats under CC BY-NC, we hope it will generate discussion and modification, and be adopted and improved upon by many research labs and graduate programs. Further, the R ‘rrchecklist’ package is licensed GPLv2 to encourage similar customized adoption.

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